Name: ____________________________________

The Sky is the Limit — Decoding Weather Conditions

Student Directions

Focus Questions: While performing this investigation, use the following focus questions to guide your work.

1. How can pressure block models of warm air and cold air be used to describe atmospheric density and pressure differences?
2. How do prevailing global winds drive the motions of hurricanes, tropical cyclones, and air masses?
3. How can locations of frontal boundaries be predicted based on data from weather maps?
4. How does latitude influence seasonal temperature gradients for fixed locations?
5. How can changes in weather conditions over three days be described using data collected from a sequence of weather maps?

Purpose of Investigation: Your tasks in **Session 1: Part 1** and **Part 2** (***Student Answer Packet 1***) will be to evaluate data and communicate information to explain how the movement and interactions of air masses result in changes in weather conditions.

Materials (per student group)**:**
- 1 set of "pressure blocks" (5 red/warm and 5 blue/cold)
- 2 index cards (3-inch x 5-inch) (labeled surface 1 and surface 2)
- 1 blue and 1 red colored pencil
- 1 calculator
- 1 ***Weather Investigation Packet***
- 1 hand lens for reading station models on maps

Use appropriate safety equipment and follow all safety procedures as determined by your teacher while performing this Investigation.

Procedure:

Session 1: Part 1

1. Write your name on ***Student Answer Packet 1***. All answers for **Part 1** will be recorded in ***Student Answer Packet 1***. Obtain one set of five red/warm "pressure blocks," one set of five blue/cold "pressure blocks," and two index cards (3-inch x 5-inch).

2. You will use the red and blue "pressure blocks" to investigate a model representing the effect of air temperature and density on atmospheric pressure.

   The blocks used in this investigation represent parcels of air. Red blocks represent warm air and blue blocks represent cold air.
   All blocks are assumed to have the same:
   - mass, regardless of their volume
   - base size
   - downward pressure exerted on the surface

3. Place <u>*one tall red block*</u> and <u>*one short blue block*</u> next to each other on their square bases on the surface of the table.

4. Place <u>*one more*</u> block of the same color on top of <u>*each*</u> block already on the table so there is a total of two blocks of each color in each stack as shown in *Figure 1*.

   Answer questions 4a and 4b.

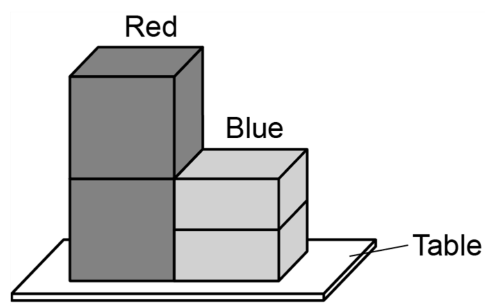

Figure 1

5. Remove <u>*one red*</u> block from the stack and place the index card that is labeled **surface 1** on top of the blocks.

6. Place <u>*two red*</u> and <u>*one blue*</u> block on top of **surface 1**, as shown in *Figure 2* (for a total of three blocks in each stack).

   Answer questions 6a, 6b, and 6c.

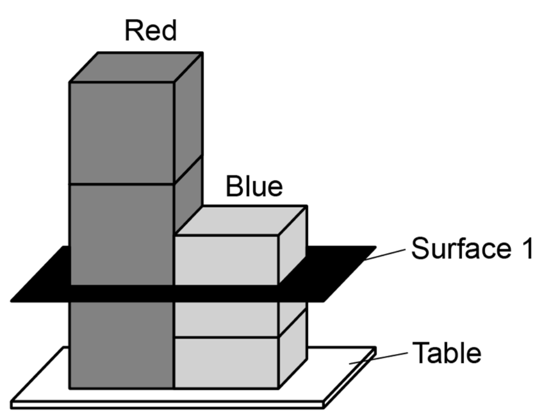

Figure 2

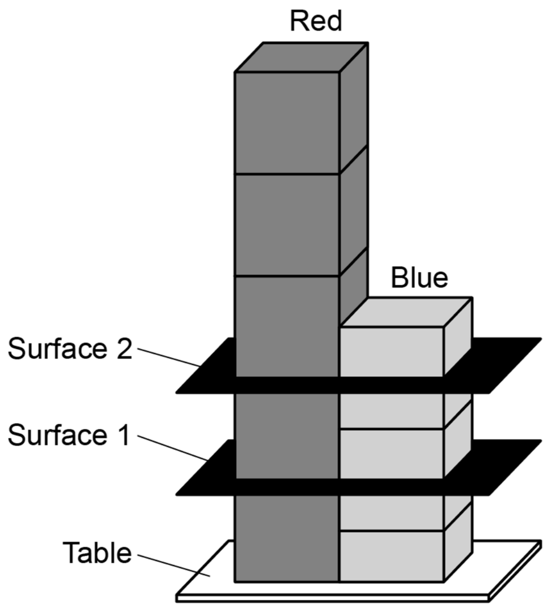

Figure 3

7. Remove _one red_ block and add one _blue block_. Then, place the index card labeled **surface 2** on the top of the stack. Add _three red_ blocks and _one blue_ block so that there is a total of five red and five blue blocks separated at **surface 1** and **surface 2**, as shown in *Figure 3*.

   Answer question 7.

8. Use the *Model of Stacked Blocks with Pressure Isolines* and the *Vertical Cross Section of Air Pressure* (*at 7 a.m. December 9, 2010*) *in a Column of Warm and Cold Air* in the ***Weather Investigation Packet*** to answer questions 8a, 8b, and 8c.

   Return the pressure block materials to your teacher.

   Weather occurs at different temporal and spatial scales. In this investigation two different meteorological scales are used, synoptic scale and global scale. Synoptic scale phenomena cover an area of several hundred to thousands of kilometers. High- and low-pressure systems shown on typical weather maps are synoptic in scale. Pressure and convection are important meteorological principles that are at the root of the development of fronts and large-scale weather systems such as hurricanes. Hurricanes occur globally in both hemispheres and are a result of pressure systems developing over tropical waters.

9. Each member of the group will plot the track of **one** hurricane. One group member will plot the track of Hurricane Andrew (with a blue pencil); the other group member will plot the track of Hurricane Orson (with a red pencil). Use the appropriate hurricane tracking chart and the data to plot the path of your selected hurricane found on pages 5 or 6 in ***Student Answer Packet 1*.** Connect your plots with a line using your colored pencil. Draw arrows along the path to indicate the direction of the hurricane.

The tables contain data for a northern hemisphere tropical system (Hurricane Andrew) and a southern hemisphere tropical cyclone (Hurricane Orson).

Data for Hurricane Andrew

Date	Time	Latitude (°N)	Longitude (°W)	Wind (mph)	Pressure (mb)	Storm Type
8/22/1992	6:00 a.m.	25.6	67.0	75	994	Category 1 hurricane
8/22/1992	6:00 p.m.	25.7	69.7	109	969	Category 2 hurricane
8/23/1992	6:00 a.m.	25.5	72.5	150	947	Category 4 hurricane
8/23/1992	6:00 p.m.	25.4	75.8	173	922	Category 5 hurricane
8/24/1992	6:00 a.m.	25.4	79.3	150	937	Category 4 hurricane
8/24/1992	6:00 p.m.	25.8	83.1	132	947	Category 4 hurricane
8/25/1992	6:00 a.m.	26.6	86.7	132	948	Category 4 hurricane
8/25/1992	6:00 p.m.	27.8	89.6	144	941	Category 4 hurricane
8/26/1992	6:00 a.m.	29.2	91.3	138	955	Category 4 hurricane
8/26/1992	6:00 p.m.	30.9	91.6	58	991	Tropical Storm
8/27/1992	6:00 a.m.	32.1	90.5	35	997	Tropical Depression
8/27/1992	6:00 p.m.	33.6	88.4	29	999	Tropical Depression

Data for Tropical Cyclone (Hurricane) Orson

Date	Time	Latitude (°S)	Longitude (°E)	Wind (mph)	Pressure (mb)	Storm Type
4/18/1989	6:00 a.m.	12.6	124.1	46	991	Category 1 hurricane
4/18/1989	6:00 p.m.	12.4	122.3	58	985	Category 1 hurricane
4/19/1989	6:00 a.m.	12.4	121.4	69	976	Category 2 hurricane
4/19/1989	6:00 p.m.	12.3	120.4	92	954	Category 3 hurricane
4/20/1989	6:00 a.m.	12.4	119.3	109	939	Category 3 hurricane
4/20/1989	6:00 p.m.	12.9	118.2	127	929	Category 4 hurricane
4/21/1989	6:00 a.m.	14.2	116.7	132	914	Category 4 hurricane
4/21/1989	6:00 p.m.	15.5	116.2	144	905	Category 5 hurricane
4/22/1989	6:00 a.m.	17.4	116.2	150	905	Category 5 hurricane
4/22/1989	6:00 p.m.	20.0	116.1	138	911	Category 4 hurricane
4/23/1989	6:00 a.m.	23.7	117.3	58	985	Category 1 hurricane

10. Compare the track that you plotted with the track plotted by your partner.

    Answer question 10.

11. Use your knowledge of global wind patterns and hurricanes to complete the task in number 11.

    Answer question 11.

Session 1: Part 2

12. Mid-latitude cyclones are characterized by frontal boundaries. Each member of the group will use information from the station models on the *Surface Observation Map* found in ***Student Answer Packet 1*** to find the center of the low-pressure system and its associated fronts. Communicate with your partner to determine the location of the center of this mid-latitude cyclone. Both members will indicate this location with a capital "L" using the red colored pencil. Determine and draw (using appropriate frontal symbols) the location of **both** the warm front and the cold front associated with this low-pressure system. Draw the warm front using the red colored pencil and draw the cold front using the blue colored pencil.

    Answer question 12.

13. Use the surface weather map in your ***Weather Investigation Packet*** from *Thursday, December 15, 2011*, to complete the task in number 13.

    Answer question 13.

14. Energy is distributed unevenly throughout Earth's atmosphere due to seasonal variations in the angle and intensity of sunlight. Determine the temperature variations between two fixed locations on Earth at different times of the year.

    Answer question 14.

15. Weather forecasts can be produced by observing daily weather conditions over periods of time. The sequence of surface weather maps in the ***Weather Investigation Packet*** shows weather data for December 14, 15, and 16, 2011.

    Use weather data from the maps to answer question 15.

Session 2

Your task in **Session 2** is to apply what you have learned in **Session 1** to independently answer the investigation questions found in ***Student Answer Packet 2***.